

Ankush Santra

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Summary

MCA student and Full-Stack Developer specializing in the MERN stack. Experienced in building scalable web applications with real-time features, secure authentication mechanisms

SKILLS

Programming Languages – C, Java, JavaScript

Frontend – HTML, CSS, Tailwind CSS, React.js, Redux Toolkit, Context API

Backend – Node.js, Express.js, Socket.io, REST APIs, JWT Authentication

Database – MongoDB, SQL

Tools - Git, GitHub, Postman

Projects

1. AI-Powered Interview Preparation (github - <https://github.com/ankushsantra42/genAI-fullstack-job-preparation>)

Full-Stack MERN | React.js, Node.js, Express.js, MongoDB, Google Gemini AI (LLM), Puppeteer, Zod, SCSS.

- Full-Stack MERN application that leverages Google Gemini AI to provide personalized interview preparation and skill gap analysis for job seekers.
- Engineered a dynamic PDF generation engine using Puppeteer to create tailored, job-specific resumes in real-time.
- Implemented Zod-based schema validation to handle complex AI responses, ensuring seamless integration between LLMs and MongoDB.
- Developed a secure session management system using JWTs and Cookie-based authentication, including a custom token-blacklist for secure sign-outs.

2. E-COMMERCE Platform (github - <https://github.com/ankushsantra42/E-commerce>)

Full-Stack MERN | React.js, Node.js, Express.js, MongoDB, Redux Toolkit, Tailwind CSS, Shadcn UI

- Built a full-stack e-commerce app with separate interfaces for user and administrators.
- Integrated secure authentication using HTTP-only cookies, JWT, and role-based routing.
- Managed complex app state and asynchronous API calls across the app using Redux Toolkit.
- Cart: Personalized shopping with cart management and real-time updates for item availability.
- Paypal Payment: Seamless online payments integrated with paypal for secure transactions.
- Created scalable RESTful APIs in Node/Express to manage users, orders, and product inventory.

3. CHAT APP (<https://chat-app-main-plly.onrender.com>)

Full-Stack MERN | React, Node.js, Express.js, MongoDB, Socket.io, Tailwind CSS, Redux Toolkit, Cloudinary.

- Architected a scalable backend with Node/Express to handle RESTful APIs alongside WebSocket connections for live chatting.
- Implemented bi-directional, low-latency communication for instant messaging and online status tracking using Socket.io.
- Ensured robust security by implementing JWT-based authentication via secure cookies and incorporating Arcjet for bot/security inspection.
- Optimized frontend performance and user experience by implementing optimistic UI updates and custom keyboard sound hooks.
- Successfully orchestrated the deployment pipeline, configuring the Express backend to serve the compiled Vite/React frontend as a single production artifact on Render.

Education

MCA | University of Calcutta (Passing Year – 2026) | SGPA-8.00

BSc in Computer Science | Vivekananda Mahavidyalaya, Haripal (Passing Year – 2024) | CGPA- 8.83

12th – Science | Nalikul Deshbandhu Bani Mandir High School (Passing Year – 2021) | Percentage: 90%

10th – General | Kinkarbati Agricultural Institution (Passing Year – 2019) | Percentage: 87%